

No. of Functional Features Included in the Solution

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features:

S.No	Functional Feature	Description	Module	Status	Contribution
1	Dataset Loading	Imports the loan applicant dataset from CSV and prepares it for processing.	Data Acquisition	Done	High
2	Data Cleaning & Transformation	Handles missing values, converts categorical attributes into machine-readable format and prepares clean data.	Data Preparation	Done	High
3	Exploratory Data Analysis	Generates statistical summaries and visualizations to understand applicant patterns and relationships.	Data Analysis	Done	Medium
4	Machine Learning Model Development	Builds and trains multiple classification models. Random Forest was selected as the final model based on performance and deployed in the Flask application.	Model Training	Done	High
5	Performance Comparison	Evaluates all trained models using accuracy score and selects the best-performing classifier.	Model Evaluation	Done	High
6	Loan Eligibility Prediction	Accepts applicant information and predicts whether the loan is likely to be approved or rejected.	Prediction Engine	Done	High
7	Web Application Development	Develops a responsive Flask-based interface for user interaction.	Web Interface	Done	Medium

8	Model Deployment	Integrates the trained Random Forest model with the Flask application for real-time prediction.	Deployment	Done	Medium
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	7
Additional / Nice-to-Have Features	1
Features Tested & Verified	8

Feature Category Breakdown:

Category	Count	Included Features	Category
User Interface	1	Loan Eligibility Input Form	User Interface
Data Processing	2	Dataset Loading, Data Cleaning	Data Processing
Machine Learning	3	Model Training, Evaluation, Prediction	Machine Learning
Deployment	1	Flask Integration	Deployment
Storage	1	CSV Dataset and Pickle Model	Storage
Authentication	0	Not Imlemented	

Key Functional Highlights

- Import and preprocess loan application data.
- Handle missing values using statistical techniques.
- Encode categorical features for machine learning.
- Train four different machine learning algorithms.
- Train and evaluate machine learning models and deploy the best-performing Random Forest classifier.
- Save the trained model using Pickle.

- Build an interactive Flask web application.
- Predict loan approval instantly based on user input.